

Experiment-9

16-Bit addition

Aim:

To write an assembly language program to implement 16-bit addition using 8086 processor.

ALGORITHM:

- 1) start the program by loading a register pair with address of 1st number.
- 2) copy the data to another register pair.
- 3) load the second number to the first register.
- 4) Add the two register to the pair contents.
- 5) check for carry.
- 6) store the value of sum & carry in memory location.

PROGRAM:

MNEUMONICS	Explanation.
MOV AX [1100H]	load 16 bit value from AX register
MOV BX [1102H]	load the 16 bit value from memory address to BX register.

ADD AX, BX

Add the value BX
to AX in AX

MOV [1200 H], AX

move the 16 bit value
in the AX register.

HLT

Halt.

INPUT:

Register	Memory	Data
AX	32	1100
BX	45	1102

Output:

Register	Memory	Data
AX	77	1200

Result:

0107 ✓

Thus the program was
executed successfully using 8086
simulator.



```
01 MOV AX, [1100H]
02 MOV BX, [1102H]
03 ADD AX, BX
04 MOV [1200H], AX
05 HLT
```

```
01 MOV AX, [1100H]
02 MOV BX, [1102H]
03 ADD AX, BX
04 MOV [1200H], AX
05 HLT
06
07
```

emulator: noname.bin_



step delay ms: 0

registers

	H	L
AX	00	08
BX	00	03
CX	00	00
DX	00	00
CS	0100	
IP	000C	
SS	0100	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0100	
ES	0100	

0100:000C

```
01000: A1 161 i
01001: 00 000 NULL
01002: 11 017 <
01003: 8B 139 i
01004: 1E 030 >
01005: 02 002 0
01006: 11 017 <
01007: 03 003 v
01008: C3 195 |
01009: A3 163 u
0100A: 00 000 NULL
0100B: 12 018 +
0100C: F4 244 r
0100D: 90 144 e
0100E: 90 144 e
0100F: 90 144 e
01010: 90 144 e
01011: 90 144 e
01012: 90 144 e
01013: 90 144 e
01014: 90 144 e
01015: 90 144 e
```

0100:000C

```
MOV AX, [01100h]
MOV BX, [01102h]
ADD AX, BX
MOV [01200h], AX
HLT
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
...
```

screen

source

reset

aux

vars

debug

stack

flags

line: 5

col: 4

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